DDD-W-631D
October 19, 1976
SUPERSEDING
Int. Fed. Spec. DDD-W-00631C (DSA-DM)
July 11, 1967 and
Fed. Spec. DDD-W-631A
April 29, 1959
(See 6.5)

FEDERAL SPECIFICATION

WRAPPERS, STERILIZATION, COTTON CLOTH, GREEN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for two types, and six sizes of a green, cotton cloth, sterilization wrapper.

1.2 Classification. The wrappers shall be furnished in the following types and sizes, as specified (see 6.2):

Type I - Wrappers with pockets, 12 by 13 inches.

Type II - Wrapper without pockets.

Size 1 - 12 inches square
Size 2 - 18 inches square
Size 3 - 24 inches square
Size 4 - 36 inches square
Size 5 - 54 inches square

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 6530

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Federal Specifications:

V-T-276	-	Thread, Cotton.
CCC-C-430	-	Cloth, Sheeting, Cotton, and Polyester and Cotton.
DDD-L-20	-	Label, for Clothing, Equipage, and Tentage (General Use).
PPP-B-636	-	Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123	-	Marking for Shipment (Civil Agencies).
Fed. Std. No. 751	-	Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Forth Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of Military Specifications, Drawings and Standards, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials and components. The materials and components shall conform to the applicable specifications and standards as specified herein.

3.1.1 Wrapper cloth. The wrappers shall be fabricated from the dyed cotton cloth conforming to CCC-C-430, type II, class 1, style A, except that the colorfastness to light is not applicable. The color of the cloth shall approximately match the grayish green 180 standard shade (see 6.3).

3.1.2 Thread. All the sewing operations shall be performed with the dyed cotton thread conforming to V-T-276, type 1A3, ticket number 30, 3 ply. The color of the thread shall approximately match the color of the dyed wrapper cloth, 3.1.1. The dyed thread shall show at least "good" colorfastness to laundering when inspected as specified in V-T-276.

3.3 Design.

3.3.1 Type I. The type I wrapper shall consist of two plies of the wrapper cloth sewn together along the sides and ends, with the ends folded onto the surface to form two pockets. The type I wrapper shall conform in appearance to figure 1.

3.3.2 Type II. The type II wrapper shall consist of two plies of the wrapper cloth sewn together along the sides and ends. The type II wrapper, as applicable, shall conform in appearance to figure 2.

3.4 Construction.

3.4.1 Seams and stitching. All the seams and stitching used in the construction of the wrappers shall conform to Fed. Std. No. 751. All sewing shall be accomplished with stitch type 301 with 10 to 12 stitches per inch. All top stitching shall be 1/16 to 1/8 inch from the edges of the wrapper. The two rows of stitching around the periphery of the type II, size 5 wrapper shall be 3/16 to 1/4 inch apart. The outer row of the stitching shall be 1/16 to 1/8 inch from the edge. The seam allowance for all wrappers except type II, size 5 shall be a minimum of 1/4 inch. The seam allowance for the type II, size 5 wrapper shall be 5/16 to 1/2 inch. The seam allowances shall be

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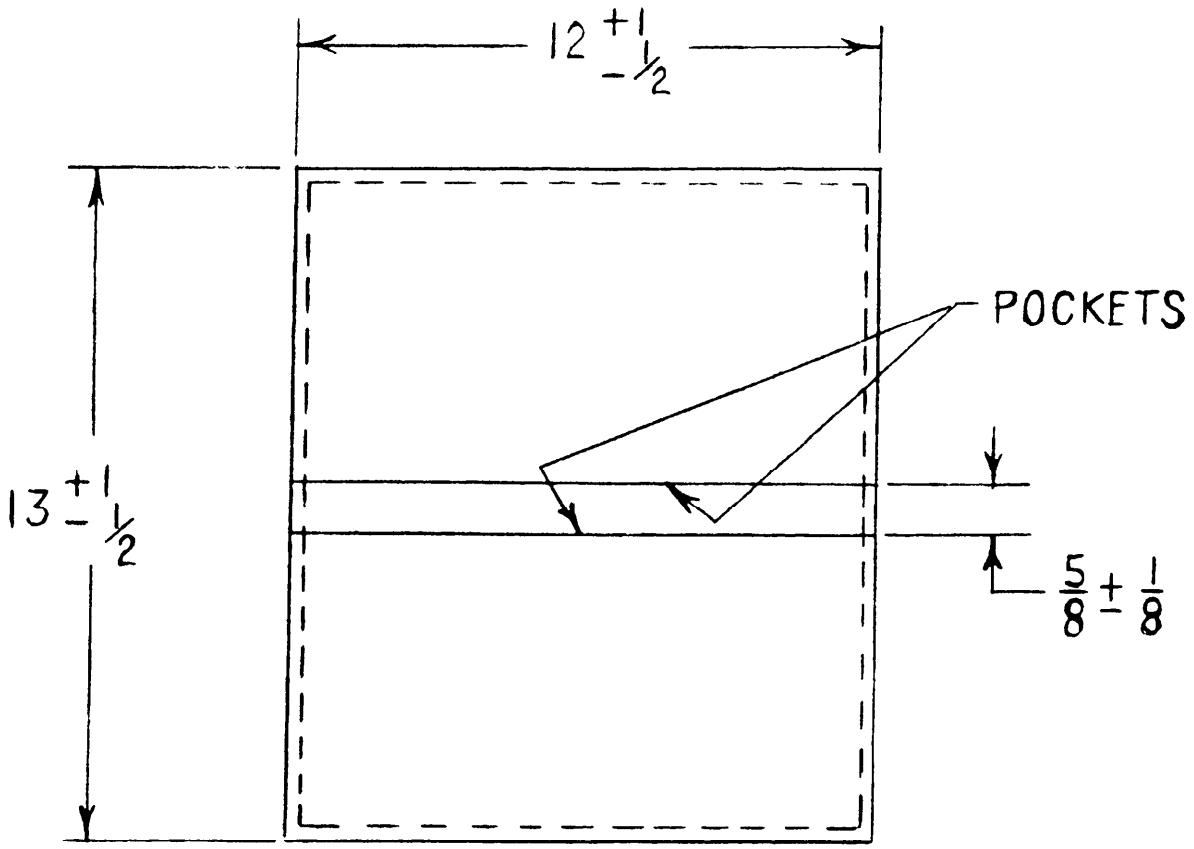
maintained so that no runoff or open seam shall occur. All the seams shall start and finish uniformly. No seam shall be twisted, puckered, or pleated. All the seam edges shall be properly forced out and shall not contain any folds. All the thread ends shall be trimmed to a maximum length of 1/4 inch. No cut or raw edge shall be visible on the exterior of the wrapper.

3.4.2 Sewing. Each row of stitching shall be straight and parallel to the seam edge. The straightness of the stitching, in any row, shall be maintained within a tolerance of plus or minus 1/32 inch. The same tolerance for spacing and parallelism shall be applicable for the double rows of stitches. The ends of the stitching that are not caught in other seams or stitching shall be backstitched, by overlapping on themselves, a minimum of 1/2 inch. Each thread break, skip, and run-off shall be overstitched not less than 1/2 inch. The thread tension shall be maintained so that there shall not be any loose or tight stitching and the lock shall be embedded in the materials sewn together.

3.4.3 Construction details.

3.4.3.1 Type I. The type I wrapper shall be fabricated from one piece of cloth folded in half and sewn together on the reverse side with seam type SSa-1 on the sides and end. One of the sides or end shall contain an opening of sufficient size to permit the wrapper to be turned rightside out. The opening shall be sewn together with seam type SSc-1. The top and bottom ends of the wrapper shall be top stitched with one row of stitching. The top and bottom ends shall then be folded onto the wrapper forming the pockets. The space between the pocket openings shall be as specified in figure 1. The sides and the top and bottom folded edges shall be top stitched with one row of stitching. As an alternate the wrapper may be fabricated from two pieces of the cloth. The two pieces of the cloth shall be superimposed on each other with all edges coinciding. The two pieces of cloth shall be sewn together on the reverse side with one row of stitches and seam type SSa-1. One of the sides or ends shall contain an opening of sufficient size to permit the wrapper to be turned rightside out. The wrapper shall then be fabricated in a similar manner as that described for the single piece of cloth folded in half.

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DIMENSIONS IN INCHES

TYPE I

FIGURE 1. Wrapper with pocket

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3.4.3.2 Type II. The type II wrappers shall be fabricated from one piece of the cloth folded in half or from two pieces of the cloth as specified in 3.4.3.1, except for the following:

a. The top and bottom ends shall not be folded onto the wrapper to form pockets.

b. Type II, sizes 1 through 4 wrappers, shall be sewn together along the sides and ends with a single row of stitching.

c. The size 5 wrapper shall be sewn together along the sides and ends with two rows of stitching (see figure 2).

3.5 Dimensions. The assembled wrappers shall conform to the dimensions for the different types and sizes specified in table I when measured as specified in 4.3.2.

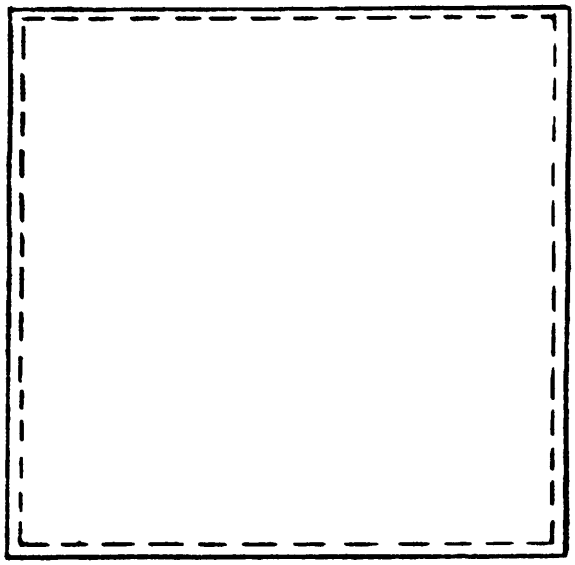
TABLE I. Dimensions (in inches)

Type	Size	Width	Length	Tolerance
I	-	12	13	+1, -1/2
II	1	12	12	+1, -1/2
II	2	18	18	± 1
II	3	24	24	± 1
II	4	36	36	± 1
II	5	54	54	± 2

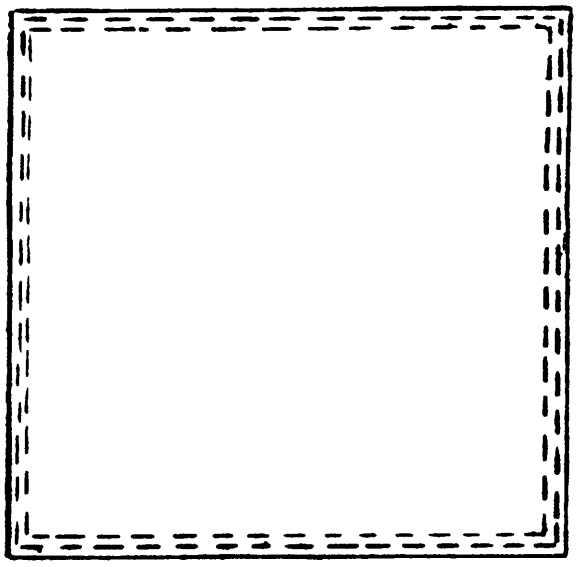
3.6 Identification marking. Each wrapper shall be legibly and distinctly marked on one of the outside corners with a nontoxic, indelible, black ink. The marking shall consist of the name or registered trademark of the manufacturer. The marking shall be in block letters. The marking shall conform to DDD-L-20, type IV, class 9, except that the colorfastness shall be the same as that specified for the class 4 label.

3.7 Workmanship. The wrappers shall not contain any spot, stain, hole, abraded area, tear, cut, mend, patch, scorch, needle chew, dust particles, lint, thread and cloth scraps or any other surface contaminant. The wrappers shall conform to the quality of grade established by this specification. The occurance of defects shall not exceed the acceptance criteria established herein.

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Sizes 1 Thru 4



Size 5

TYPE II

FIGURE 2. Wrapper without pocket

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsibile for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Inspection lot. An inspection lot size shall be expressed in units of one wrapper made essentially under the same conditions and from the same materials and components. The sample unit for inspection shall be one wrapper.

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4.2.2 For examination. The sample size, acceptance criteria, examinations and tests required for the wrappers or the preparation for delivery, as applicable, shall be as specified in table II and MIL-STD-105.

TABLE II Sampling for examination

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	4.0
Total defects (Major and Minor combined)	II	10.0
For dimensional examination	S-3	4.0

4.3 Inspection methods.

4.3.1 Visual examination.

4.3.1.1 Wrappers. Each wrapper selected in accordance with table II as a sample unit from the lot shall be thoroughly checked dimensionally and examined visually to determine conformance to this specification. The classification of defects, table III, shall be used to classify and enumerate the defects found.

TABLE III. Classification of defects

Defect	Major	Minor
GENERAL		
a. Any hole, scissors or knife cut, tear, mend, burn, or weakening defect such as smash, multiple float, loose slub, needle chew, or abraded area.	X	
b. Any portion of the wrapper stiffened or scorched by any process of manufacturing.	X	
c. Any missing or broken yarn, thin place, misweave, woven in waste, loose yarn, snagged thread, area of no dye penetration, or dye streak on the exterior of the wrapper.	1/	

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TABLE III. Classification of defects (cont'd)

Defect	Major	Minor
SHADED PARTS		
a. Any part shaded or any shade bar.		X
b. Color of any component not as specified or not uniform.		X
CLEANNESS		
a. Three or more thread ends not trimmed to a maximum length of 1/4 inch.		X
b. Any thread or cloth scraps not removed or contains any lint, dust particles or any other surface contaminant.		X
c. Any unsightly slub, spot, or stain.		X
COMPONENTS AND ASSEMBLY		
a. Any component not as specified or any defect of component or defect of assembly, not herein classified.	1/	
b. Any component, component part, or required operation omitted, or any operation improperly performed, not herein classified.	1/	
SEAMS AND STITCHING		
a. Accuracy of seaming:		
1. Any seam twisted, puckered, or pleated; any part of the wrapper caught in an unrelated operation or stitching; any seam edge not properly forced out or contains any fold.		X

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TABLE III. Classification of defects (cont'd)

Defects	Major	Minor
2. Ends of stitching that are not caught in other seams or stitching:		
(a) Any not securely backstitched.	X	
(b) Any secure but backstitched for a distance less than 1/2 inch or the backstitching is not overlapped on itself.		X
3. Thread breaks:		
(a) Any not securely overstitched.	X	
(b) Any secured but overstitched for a distance less than 1/2 inch.		X
b. Gage of stitching 2/;		
1. Any stitching irregular, uneven or not gaged as specified, or not uniform.		X
c. Open seams 3/:		
1. Any more than 1/8 inch but not more than 1 inch.		X
2. Any more than 1 inch.	X	
d. Raw edges 4/:		
1. Any more than 1/4 inch but not more than 1 inch.		X
2. Any more than 1 inch.	X	
e. Seam and stitch type:		
1. Not specified seam or stitch.	1/	
f. Stitch tension:		
1. Any loose stitching 5/.	1/	
2. Any tight stitching resulting in breakage of the stitches when normal pull is applied 6/.	X	

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TABLE III. Classification of defects (cont'd)

Defect	Major	Minor
g. Stitches per inch 7/:		
1. Stitches more or less than specified:		
(a) One or 2.		X
(b) More than 2.	X	
MARKING		
a. Missing, incorrect or illegible.		X
b. Misplaced.		X

1/ The defect shall be classified and scored as a major defect, when it seriously affects the serviceability or appearance, otherwise it shall be classified and scored as a minor defect.

2/ The gage of stitching defect shall be scored only when the condition exists for more than 2 inches on any row of stitching.

3/ A seam shall be classified and scored as open when one or more stitches joining a seam are broken or when two or more continuous skipped stitches or run-offs occur. On double stitched seams, a seam shall be classified and scored as open when either one or both sides of the seam contain any of the aforementioned for a single stitched seam.

4/ Any edge that should be turned under and is not, but is securely caught in the stitching, shall be classified and scored as a raw edge. Any raw edge, that is not securely caught in the stitching, shall be classified and scored as an open seam.

5/ A seam shall be classified and scored as loose when the top and bobbin threads are on the same surface.

6/ Puckering is evidence of tight tension or gathering of the material. When puckering is evident and is not caused by the gathering of the material, the stitching shall be inspected for tight tension and breaking of stitches by exerting normal pull in the lengthwise direction of the stitching, by pulling the material taut to straighten out the seam. Puckering due to the gathering of the material shall be classified and scored as an accuracy of seaming defect.

7/ The stitches per inch defect shall be scored only when the condition exists for more than 3 inches on any row of stitching.

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4.3.1.2 Dimensional examination. The wrappers shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.3.2 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

TABLE IV. Examination of preparation for delivery

Examine	Defect
Contents	not as specified.
Container	not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.3.2.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

4.3.2 Procedure for measuring. The wrapper to be measured shall be laid flat, without tension, on a smooth, flat surface so that any wrinkles or creases shall not affect the measurement. The measurement shall be made with the use of any suitable measuring device.

4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Unit package is not required.

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5.1.1.2 Intermediate package. Each wrapper shall be compactly folded and packaged in quantities as specified in table V in a close fitting box conforming to PPP-B-636, type CF, class domestic. Closure of box shall be adequate to prevent accidental opening. Intermediate package is not required for type II, size 5 wrapper.

5.1.2 Level C. Wrappers shall be packaged in standard commercial containers of the size and kind commonly used which will afford the degree of protection required for shipment and use of the product for its intended purpose.

TABLE V. Packaging and packing quantities

Type	Size	Quantity per intermediate package	Quantity per exterior container
I	-	6	360
II	1	6	600
II	2	6	432
II	3	6	240
II	4	6	96
II	5	None	30

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Wrappers shall be packed in quantities as specified in table V in an exterior container designed for a type 2 load and consturcted in accordance with PPP-B-636, class water resistant, grade V3c. Closure, waterproofing and strapping shall be as specified in the appendix of PPP-B-636.

5.2.2 Level B. Wrappers shall be packed in quantities as specified in table V in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of PPP-B-636.

5.2.3 Level C. Wrappers shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities. Each intermediate package and exterior container shall be marked as specified in MIL-STD-129.

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6. NOTES

6.1 Intended use. The wrappers covered by this specification are intended for use in holding surgeons gloves and other surgical supplies during the sterilization process and in maintaining sterility for short periods of time.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. Type and size required (see 1.2).
c. Selection of applicable level of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample for shade. The standard sample for shade will be furnished by the contracting officer.

6.4 Catalog information. This specification is intended to cover the following items listed in the Federal Supply Catalog:

National Stock Number	Item Identification	Type
6530-00-197-9228	Wrapper, Sterilization, Green, 54 inches square, each	II
6530-00-299-9603	Wrapper, Sterilization, Green, 12 by 13 inches (pocket type), each	I
6530-00-926-4902	Wrapper, Sterilization, Green, 12 inches square, each	II
6530-00-926-4903	Wrapper, Sterilization, Green, 18 inches square, each	II
6530-00-926-4904	Wrapper, Sterilization, Green, 24 inches square, each	II
6530-00-926-4905	Wrapper, Sterilization, Green, 36 inches square, each	II

6.5 Cross-reference of classification. The changes in classification of the sterilization wrappers between this specification DDD-W-631D and superseded specifications DDD-W-00631c (DSA-DM) and DDD-W-631a are as follows:

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DDD-W-631A	DDD-W-00631c	DDD-W-631D
Type I	Type I, Style A	Type I,
--	Type I, Style B	--
Type II	Type II	Type II
Size 1	Size 1	Size 1
Size 2	Size 2	Size 2
Size 3	Size 3	Size 3
Size 4	Size 4	Size 4
--	Size 5	--
--	Size 6	Size 5
--	Size 7	--
--	Class 1	--
--	Class 2	--

MILITARY CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
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Civil Agency Coordinating Activities
VA-DMS
GSA-FSS

Project No. 6530-1363

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DOCUMENT IDENTIFIER AND TITLE
DDD-W-631D WRAPPERS, STERILIZATION, COTTON, GREEN

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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